

RetailBank Data Quality & Reject Report

Batches: day1-baseline and day2-incremental | Source: Retailbank_SourceData | Pipeline run: 6 October 2026

1,679	1,291	388	23.1%	134	15
Source rows received	Rows passed to Silver & Gold	Rows quarantined	Overall reject rate	Transaction DQ rejects*	Repair / flag categories

* Excludes MISSING_DIM, consistent with the source report.

Executive Summary

The pipeline processed 1,679 rows across the two reporting batches. A total of 1,291 rows passed validation and were loaded to Silver and Gold, while 388 rows were quarantined. The overall reject rate was 23.1%. Day 1 contained the majority of data-quality issues, particularly in transactions, while the Day 2 incremental dimension files passed without rejects. Transaction rejects remained comparatively high in both batches.

Key observation

Transactions are the main source of data-quality failures. The largest recurring issue is MISSING_DIM, followed by conflicting duplicates and invalid transaction amounts.

1. Outcome by File and Batch

Batch	Entity	Source file	Received	Passed	Rejected	Reject %	Top reasons
day1-baseline	branches	branches.csv	102	97	5	4.90%	CONFLICTING_DUPLICATE=4; DUPLICATE_RECORD=1
day1-baseline	customers	customers.csv	257	240	17	6.61%	INVALID_DOB=6; DUPLICATE_RECORD=6; MALFORMED_STRUCTURE=4
day1-baseline	products	products_json.txt	114	105	9	7.89%	CONFLICTING_DUPLICATE=6; NEGATIVE_PRICE=3
day1-baseline	transactions	transactions.csv	940	647	293	31.17%	MISSING_DIM=185; CONFLICTING_DUPLICATE=32; AMOUNT_BLANK=22
day2-incremental	branches	branches_2.csv	2	2	0	0.00%	—
day2-incremental	customers	customer_updates_2.csv	47	47	0	0.00%	—
day2-incremental	products	products_2_json.txt	18	18	0	0.00%	—
day2-incremental	transactions	transactions_2.csv	199	135	64	32.16%	MISSING_DIM=38; CONFLICTING_DUPLICATE=6; AMOUNT_ZERO=5

Control point

Received = passed + rejected for every file. Rejected rows retain their raw payload in quarantine/, so no row is dropped silently.

2. Rejects by Batch, Entity and Error Code

Batch	Entity	Error code	Rows	How it was handled
day1-baseline	branches	CONFLICTING_DUPLICATE	4	Quarantined (all copies). Same key, different values; the correct value cannot be determined.

Batch	Entity	Error code	Rows	How it was handled
day1-baseline	branches	DUPLICATE_RECORD	1	Quarantined (extra copies only). Best copy kept.
day1-baseline	customers	ACCOUNT_CONFLICT	1	Quarantined. Account shared by two customers; customer whose ID matches the account is kept.
day1-baseline	customers	DUPLICATE_RECORD	6	Quarantined (extra copies only). Best copy kept.
day1-baseline	customers	INVALID_DOB	6	Quarantined. DOB is in the future, under 18, or over 110.
day1-baseline	customers	MALFORMED_STRUCTURE	4	Quarantined. Required key (customer, account, or branch name) is missing.
day1-baseline	products	CONFLICTING_DUPLICATE	6	Quarantined (all copies). Same key, different values; the correct value cannot be determined.
day1-baseline	products	NEGATIVE_PRICE	3	Quarantined. Not auto-corrected; needs a source fix.
day1-baseline	transactions	AMOUNT_BLANK	22	Quarantined. Amount missing or non-numeric.
day1-baseline	transactions	AMOUNT_NEGATIVE	16	Quarantined. Negative amount.
day1-baseline	transactions	AMOUNT_ZERO	14	Quarantined. Zero amount.
day1-baseline	transactions	CONFLICTING_DUPLICATE	32	Quarantined (all copies). Same key, different values; the correct value cannot be determined.
day1-baseline	transactions	DUPLICATE_RECORD	20	Quarantined (extra copies only). Best copy kept.
day1-baseline	transactions	FUTURE_DATE	4	Quarantined. Date or timestamp is after the run date.
day1-baseline	transactions	MISSING_DIM	185	Quarantined. Key not in the dimension; excluded from the reject threshold. Reprocess once the dimension arrives.
day2-incremental	transactions	AMOUNT_BLANK	3	Quarantined. Amount missing or non-numeric.
day2-incremental	transactions	AMOUNT_NEGATIVE	3	Quarantined. Negative amount.
day2-incremental	transactions	AMOUNT_ZERO	5	Quarantined. Zero amount.
day2-incremental	transactions	CONFLICTING_DUPLICATE	6	Quarantined (all copies). Same key, different values; the correct value cannot be determined.
day2-incremental	transactions	DUPLICATE_RECORD	4	Quarantined (extra copies only). Best copy kept.
day2-incremental	transactions	FUTURE_DATE	5	Quarantined. Date or timestamp is after the run date.
day2-incremental	transactions	MISSING_DIM	38	Quarantined. Key not in the dimension; excluded from the reject threshold. Reprocess once the dimension arrives.

3. Rejects by Rule

Rule	Error code	Rows	Meaning
CU-ACCT	ACCOUNT_CONFLICT	1	
TX-AMT-BLANK	AMOUNT_BLANK	25	
TX-AMT-NEG	AMOUNT_NEGATIVE	19	
TX-AMT-ZERO	AMOUNT_ZERO	19	
BR-CONFLICT	CONFLICTING_DUPLICATE	4	
PR-CONFLICT	CONFLICTING_DUPLICATE	6	
TX-CONFLICT	CONFLICTING_DUPLICATE	38	
BR-DUP	DUPLICATE_RECORD	1	
CU-DUP	DUPLICATE_RECORD	6	
TX-DUP	DUPLICATE_RECORD	24	
TX-FUT-DATE	FUTURE_DATE	9	

Rule	Error code	Rows	Meaning
CU-DOB	INVALID_DOB	6	
CU-STRUCT	MALFORMED_STRUCTURE	4	
TX-RI-ACCT	MISSING_DIM	57	account_id not in dim_customer
TX-RI-BR	MISSING_DIM	64	branch_id not in dim_branch
TX-RI-PROD	MISSING_DIM	102	product_id not in dim_product
PR-PRICE-NEG	NEGATIVE_PRICE	3	

4. Repaired and Flagged Data

The following records were loaded rather than rejected. Repairs and flags are retained for auditability.

Entity	Flag	Rows	How it was handled
transactions	DATE_RECONCILED_TO_TIMESTAMP	214	Date column disagreed with timestamp; timestamp used as truth. Logged.
transactions	DATE_AMBIGUOUS_RESOLVED_BY_TS	65	Slash date readable as DD/MM or MM/DD; timestamp selected the interpretation.
transactions	IS_REFUND_DEFAULTED	110	is_refund was blank; defaulted to false.
transactions	NON_INR_CURRENCY	17	USD/EUR retained; excluded from INR value KPIs. Not converted.
transactions	LATE_CORRECTION	9	Day 2 correction applied in place to the existing fact row.
customers	DOB_MISSING	5	DOB blank; retained with null DOB.
customers	EMAIL_MISSING	5	Email blank; retained with null email.
customers	EMAIL_INVALID	4	Malformed email set to null after fixing case and '@@'.
customers	PHONE_INVALID	3	Phone not parseable as 10 digits; set to null.
customers	REG_DATE_AMBIGUOUS_ASSUMED_DMY	3	Registration date readable both ways; day-first assumed.
branches	MANAGER_MISSING	4	Manager blank; set to UNASSIGNED.
branches	PHONE_INVALID	1	Phone invalid; set to null.
branches	MANAGER_LABEL_STRIPPED	1	'(New)' label removed from the manager name.
products	IS_ACTIVE_DEFAULTED	3	is_active missing or null; defaulted to false.
products	JSON_COMMA_REPAIRED	1	products_json.txt missing a comma between objects; file repaired before parsing (file-level event).

Auditability

Flagged rows load into Silver and Gold. The flag is stored in dq_flags, allowing each repair to be traced after loading.

5. Threshold and Conflict Policy

Reject threshold

The configured reject threshold is 25% per entity, excluding MISSING_DIM. The highest rate excluding MISSING_DIM is transactions on Day 1 at approximately 11.5%, so no batch breached the threshold. The threshold remains a placeholder pending data-owner sign-off.

Conflicting duplicates

When a key has different values across copies, every copy is quarantined rather than guessing which value is correct. This affects 48 rows across both batches, including product IDs P002, P063 and P089. Transactions pointing to quarantined products or branches can subsequently appear as MISSING_DIM.

Late corrections

Day 2 corrections that pass validation are applied in place to existing fact rows. Correction T0793 has a 2027 date, so it is quarantined as FUTURE_DATE and does not change the fact. The other four Day 2 FUTURE_DATE rejects are new transactions dated in 2027.

6. Open Items and Recommended Actions

Priority	Action	Reason
1	Confirm the 25% reject threshold with data owners.	The current threshold is explicitly marked as a placeholder.
2	Fix the 3 negative product prices at source.	These records cannot be safely auto-corrected.
3	Resolve the 223 orphan transaction keys.	Missing dimension keys are the largest recurring issue and require dimension/source alignment.
4	Resolve conflicting duplicates.	Conflicting values cannot be safely selected by the pipeline.
5	Confirm that day-first is the correct interpretation for ambiguous dates.	The current treatment is an assumption based on available timestamp information.
6	Confirm treatment of non-INR transaction amounts.	The current process excludes them from INR value KPIs rather than converting them.

Overall assessment

The pipeline provides a controlled path for both clean and problematic records: valid data continues downstream, rejected rows are quarantined, and repaired rows are explicitly flagged. The immediate focus should be on transaction-level dimension integrity and source-system duplicate/amount issues.